

EXTENSION 2

Stage 6

Proof (Ext2), P1 Nature of Proof (Ext2)

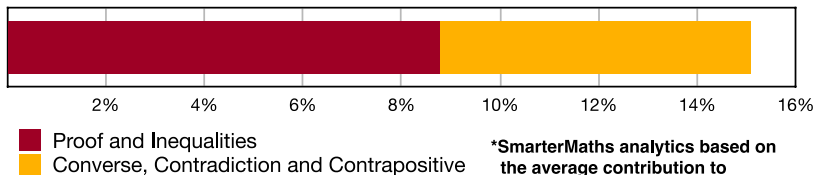
Proof and Inequalities

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Exam Equivalent Time: 132 minutes (based on allocation of 1.5 minutes per mark)



P1 Nature of Proof



HISTORICAL CONTRIBUTION

- P1 Nature of Proof* has contributed a surprising average of 15.0% per new syllabus Ext2 exam since it was introduced in 2020.
- This topic has been split into two categories for analysis purposes: *1-Proof and Inequalities (8.6%)* and *2-Converse, Contradiction and Contrapositive Proof (6.4%)*.
- This analysis looks at *Proof and Inequalities*.

HSC ANALYSIS - What to expect and common pitfalls

- Proof and Inequalities (8.6%)* is one of the silverbacks in the Extension 2 jungle that again beat its chest in 2023.
- For the third year in a row, this topic area attracted a multiple choice and 2 longer answer questions. The new syllabus mark allocations to this topic area have surprised to the upside and look set to continue, in our view.
- Question difficulty has been across the whole spectrum, ranging from band 2 to band 6 in just the new syllabus era.
- A common question type requires students to use the property that squares of real numbers are non-negative, asked in moderate (2023 Ext2 12b) to low difficulty examples (2022 Ext2 12a and 2020 Ext2 13c) in the last four years.
- Proofs using a given equation appeared in substantial 4-mark questions in 2023 Ext2 16b, 2022 Ext2 16c and 2021 Ext2 15a. Students found each question very challenging and revision attention is highly recommended.
- Triangle inequality proofs were required in 2022 Ext2 15d and 2021 within a cross-topic vector themed example.
- The database contains a rich history of past Ext2 inequality proof questions that cover many proofs utilising $a > b$ and arithmetic/geometric mean properties on top of the above examples highlighted.

Questions

1. Proof, EXT2 P1 2018 HSC 9 MC

It is given that a, b are real and p, q are purely imaginary.

Which pair of inequalities must always be true?

- $a^2p^2 + b^2q^2 \leq 2abpq$, $a^2b^2 + p^2q^2 \leq 2abpq$
- $a^2p^2 + b^2q^2 \leq 2abpq$, $a^2b^2 + p^2q^2 \geq 2abpq$
- $a^2p^2 + b^2q^2 \geq 2abpq$, $a^2b^2 + p^2q^2 \leq 2abpq$
- $a^2p^2 + b^2q^2 \geq 2abpq$, $a^2b^2 + p^2q^2 \geq 2abpq$

2. Proof, EXT2 P1 2017 HSC 10 MC

Suppose $f(x)$ is a differentiable function such that

$$\frac{f(a) + f(b)}{2} \geq f\left(\frac{a+b}{2}\right), \text{ for all } a \text{ and } b.$$

Which statement is always true?

- $\int_0^1 f(x) \, dx \geq \frac{f(0) + f(1)}{2}$
- $\int_0^1 f(x) \, dx \leq \frac{f(0) + f(1)}{2}$
- $f'\left(\frac{1}{2}\right) \geq 0$
- $f'\left(\frac{1}{2}\right) \leq 0$

3. Proof, EXT2 P1 2021 HSC 5 MC

Which of the following statements is FALSE?

- $\forall a, b \in \mathbb{R}$, $a < b \Rightarrow a^3 < b^3$
- $\forall a, b \in \mathbb{R}$, $a < b \Rightarrow e^{-a} > e^{-b}$
- $\forall a, b \in (0, +\infty)$, $a < b \Rightarrow \ln a < \ln b$
- $\forall a, b \in \mathbb{R}$, with $a, b \neq 0$, $a < b \Rightarrow \frac{1}{a} > \frac{1}{b}$

4. Proof, EXT2 P1 2022 HSC 12a

For real numbers $a, b \geq 0$ prove that $\frac{a+b}{2} \geq \sqrt{ab}$. (2 marks)

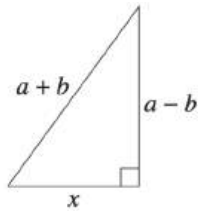
5. Proof, EXT2 P1 2017 HSC 13a

Show that $\frac{r+s}{2} \geq \sqrt{rs}$ for $r \geq 0$ and $s \geq 0$. (1 mark)

6. Proof, EXT2 P1 2020 HSC 13c

i. By considering the right-angled triangle below, or otherwise, prove that

$$\frac{a+b}{2} \geq \sqrt{ab}, \text{ where } a > b \geq 0. \quad (2 \text{ marks})$$



ii. Prove that $p^2 + 4q^2 \geq 4pq$. (1 marks)

7. Proof, EXT2 P1 2023 HSC 12b

Prove that for all real numbers x and y , where $x^2 + y^2 \neq 0$,

$$\frac{(x+y)^2}{x^2+y^2} \leq 2. \quad (2 \text{ marks})$$

8. Proof, EXT2 P1 2006 HSC 8a

Suppose $0 \leq t \leq \frac{1}{\sqrt{2}}$.

i. Show that $0 \leq \frac{2t^2}{1-t^2} \leq 4t^2$. (2 marks)

ii. Hence show that $0 \leq \frac{1}{1+t} + \frac{1}{1-t} - 2 \leq 4t^2$. (1 mark)

iii. By integrating the expressions in the inequality in part (ii) with respect to t from $t = 0$ to $t = x$ (where $0 \leq x \leq \frac{1}{\sqrt{2}}$), show that

$$0 \leq \log_e \left(\frac{1+x}{1-x} \right) - 2x \leq \frac{4x^3}{3}. \quad (2 \text{ marks})$$

iv. Hence show that for $0 \leq x \leq \frac{1}{\sqrt{2}}$

$$1 \leq \left(\frac{1+x}{1-x} \right) e^{-2x} \leq e^{\frac{4x^3}{3}}. \quad (1 \text{ mark})$$

9. Proof, EXT2 P1 2015 HSC 15c

For positive real numbers x and y , $\sqrt{xy} \leq \frac{x+y}{2}$. (Do NOT prove this.)

i. Prove $\sqrt{xy} \leq \sqrt{\frac{x^2+y^2}{2}}$, for positive real numbers x and y . (1 mark)

ii. Prove $\sqrt[4]{abcd} \leq \sqrt{\frac{a^2+b^2+c^2+d^2}{4}}$, for positive real numbers a, b, c and d . (2 marks)

10. Proof, EXT2 P1 SM-Bank 6

If $x, y, z \in \mathbb{R}$ and $x \neq y \neq z$, then

i. Prove $x^2 + y^2 + z^2 > yz + zx + xy$ (2 marks)

ii. If $x + y + z = 1$, show $yz + zx + xy < \frac{1}{3}$ (2 marks)

11. Proof, EXT2 P1 2011 HSC 5b

If p, q and r are positive real numbers and $p + q \geq r$, prove that

$$\frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \geq 0. \quad (3 \text{ marks})$$

12. Proof, EXT2 P1 2016 HSC 14c

Show that $x\sqrt{x} + 1 \geq x + \sqrt{x}$, for $x \geq 0$. (3 marks)

13. Proof, EXT2 P1 2022 HSC 15d

The complex number z satisfies $\left| z - \frac{4}{z} \right| = 2$.

Using the triangle inequality, or otherwise, show that $|z| \leq \sqrt{5} + 1$. (3 marks)

14. Proof, EXT2 P1 2023 HSC 16b

i. Prove that $x > \ln x$, for $x > 0$. (2 marks)

ii. Using part (i), or otherwise, prove that for all positive integers n ,

$$e^{n^2+n} > (n!)^2. \quad (3 \text{ marks})$$

15. Proof, EXT2 P1 2007 HSC 7a

i. Show that $\sin x < x$ for $x > 0$. (2 marks)

ii. Let $f(x) = \sin x - x + \frac{x^3}{6}$.

Show that the graph of $y = f(x)$ is concave up for $x > 0$. (2 marks)

iii. By considering the first two derivatives of $f(x)$, show that $\sin x > x - \frac{x^3}{6}$ for $x > 0$. (2 marks)

16. Proof, EXT2 P1 2012 HSC 15a

- i. Prove that $\sqrt{ab} \leq \frac{a+b}{2}$, where $a \geq 0$ and $b \geq 0$. (1 mark)
- ii. If $1 \leq x \leq y$, show that $x(y-x+1) \geq y$. (2 marks)
- iii. Let n and j be positive integers with $1 \leq j \leq n$.

Prove that $\sqrt{n} \leq \sqrt{j(n-j+1)} \leq \frac{n+1}{2}$. (2 marks)

- iv. For integers $n \geq 1$, prove that

$$(\sqrt{n})^n \leq n! \leq \left(\frac{n+1}{2}\right)^n. \quad (1 \text{ mark})$$

17. Proof, EXT2 P1 2015 HSC 15b

Suppose that $x \geq 0$ and n is a positive integer.

- i. Show that $1-x \leq \frac{1}{1+x} \leq 1$. (2 marks)
- ii. Hence, or otherwise, show that

$$1 - \frac{1}{2n} \leq n \ln \left(1 + \frac{1}{n}\right) \leq 1. \quad (2 \text{ marks})$$

- iii. Hence, explain why

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e. \quad (1 \text{ mark})$$

18. Vectors, EXT2 V1 2021 HSC 16a

- i. The point $P(x, y, z)$ lies on the sphere of radius 1 centred at the origin O .

Using the position vector of P , $\overrightarrow{OP} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, and the triangle inequality, or otherwise, show that $|x| + |y| + |z| \geq 1$. (2 marks)

- ii. Given the vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, show that

$$|a_1b_1 + a_2b_2 + a_3b_3| \leq \sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}. \quad (3 \text{ marks})$$

- iii. As in part (i), the point $P(x, y, z)$ lies on the sphere of radius 1 centred at the origin O .

Using part (ii), or otherwise, show that $|x| + |y| + |z| \leq \sqrt{3}$. (2 marks)

19. Proof, EXT2 P1 2012 HSC 16c

Let n be an integer where $n > 1$. Integers from 1 to n inclusive are selected randomly one by one with repetition being possible. Let $P(k)$ be the probability that exactly k different integers are selected before one of them is selected for the second time, where $1 \leq k \leq n$.

- i. Explain why $P(k) = \frac{(n-1)!k}{n^k(n-k)!}$. (2 marks)

- ii. Suppose $P(k) \geq P(k-1)$. Show that $k^2 - k - n \leq 0$. (2 marks)

- iii. Show that if $\sqrt{n + \frac{1}{4}} > k - \frac{1}{2}$ then the integers n and k satisfy $\sqrt{n} > k - \frac{1}{2}$. (2 marks)

- iv. Hence show that if $4n + 1$ is not a perfect square, then $P(k)$ is greatest when k is the closest integer to $\sqrt{n}$.

You may use part (iii) and also that $k^2 - k - n > 0$ if $P(k) < P(k-1)$. (2 marks)

20. Proof, EXT2 P1 2013 HSC 16a

- i. Find the minimum value of $P(x) = 2x^3 - 15x^2 + 24x + 16$, for $x \geq 0$. (2 marks)
- ii. Hence, or otherwise, show that for $x \geq 0$,

$$(x+1)\left(x^2 + (x+4)^2\right) \geq 25x^2. \quad (1 \text{ mark})$$

- iii. Hence, or otherwise, show that for $m \geq 0$ and $n \geq 0$,

$$(m+n)^2 + (m+n+4)^2 \geq \frac{100mn}{m+n+1}. \quad (2 \text{ marks})$$

21. Proof, EXT2 P1 2021 HSC 15a

For all non-negative real numbers x and y , $\sqrt{xy} \leq \frac{x+y}{2}$. (Do NOT prove this.)

- i. Using this fact, show that for all non-negative real numbers a, b and c ,

$$\sqrt{abc} \leq \frac{a^2 + b^2 + 2c}{4}. \quad (2 \text{ marks})$$

- ii. Using part (i), or otherwise, show that for all non-negative real numbers a, b and c ,

$$\sqrt{abc} \leq \frac{a^2 + b^2 + c^2 + a + b + c}{6}. \quad (2 \text{ marks})$$

22. Proof, EXT2 P1 2021 HSC 15b

For integers $n \geq 1$, the triangular numbers t_n are defined by $t_n = \frac{n(n+1)}{2}$, giving $t_1 = 1, t_2 = 3, t_3 = 6, t_4 = 10$ and so on.

For integers $n \geq 1$, the hexagonal numbers h_n are defined by $h_n = 2n^2 - n$, giving $h_1 = 1, h_2 = 6, h_3 = 15, h_4 = 28$ and so on.

- i. Show that the triangular numbers t_1, t_3, t_5 , and so on, are also hexagonal numbers. (2 marks)
- ii. Show that the triangular numbers t_2, t_4, t_6 , and so on, are not hexagonal numbers. (1 mark)

23. Proof, EXT2 P1 2022 HSC 16c

It is given that for positive numbers $x_1, x_2, x_3, \dots, x_n$ with arithmetic mean A ,

$$\frac{x_1 \times x_2 \times x_3 \times \dots \times x_n}{A^n} \leq 1 \quad (\text{Do NOT prove this.})$$

Suppose a rectangular prism has dimensions a, b, c and surface area S .

i. Show that $abc \leq \left(\frac{S}{6}\right)^{\frac{3}{2}}$. (2 marks)

ii. Using part (i), show that when the rectangular prism with surface area S is a cube, it has maximum volume. (2 marks)

24. Proof, EXT2 P1 2014 HSC 15a

Three positive real numbers a, b and c are such that $a + b + c = 1$ and $a \leq b \leq c$.

By considering the expansion of $(a + b + c)^2$, or otherwise, show that

$$5a^2 + 3b^2 + c^2 \leq 1. \quad (2 \text{ marks})$$

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Worked Solutions

1. Proof, EXT2 P1 2018 HSC 9 MC

$a, b \rightarrow$ real $p, q \rightarrow$ purely imaginary
 $\Rightarrow ab, pq, (ab-pq)$ are real
 $\Rightarrow ap, bq, (ap-bq)$ are purely imaginary.

$$(ab-pq)^2 \geq 0$$

$$a^2b^2 + p^2q^2 \geq 2abpq \quad (\text{Eliminate A and C})$$

$$(ap-bq)^2 \leq 0$$

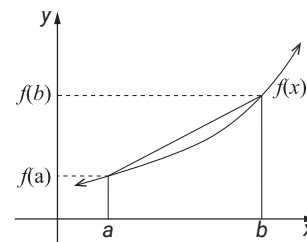
$$a^2p^2 + b^2q^2 \leq 2abpq$$

$$\Rightarrow B$$

2. Proof, EXT2 P1 2017 HSC 10 MC

Representing the equation graphically
 (one of many possibilities)

♦ Mean mark 41%.



$f(x)$ is an increasing function between a and b such that:

$$\frac{f(a) + f(b)}{2} \geq f\left(\frac{a+b}{2}\right)$$

If $a = 0$ and $b = 1$,

$$\int_0^1 f(x) dx \leq \text{Area of trapezium}$$

$$\leq \frac{1}{2}(1)(f(0) + f(1))$$

$$\leq \frac{f(0) + f(1)}{2}$$

$\Rightarrow B$

3. Proof, EXT2 P1 2021 HSC 5 MC

By contradiction:

Consider D

Let $a = -1$ and $b = 1$,

$$a < b \rightarrow -1 < 1 \text{ (TRUE)}$$

$$\frac{1}{a} > \frac{1}{b} \rightarrow -1 > 1 \text{ (FALSE)}$$

$\Rightarrow D$ is false

♦ Mean mark 50%.

4. Proof, EXT2 P1 2022 HSC 12a

Since $(\sqrt{a} - \sqrt{b})^2 \geq 0$:

$$a - 2\sqrt{ab} + b \geq 0$$

$$a + b \geq 2\sqrt{ab}$$

$$\therefore \frac{a+b}{2} \geq \sqrt{ab} \dots \text{as required}$$

Mean mark 93%.

5. Proof, EXT2 P1 2017 HSC 13a

$$(\sqrt{r} - \sqrt{s})^2 \geq 0$$

$$r - 2\sqrt{r}\sqrt{s} + s \geq 0$$

$$r + s \geq 2\sqrt{r}\sqrt{s}$$

$$\frac{r+s}{2} \geq \sqrt{rs}$$

6. Proof, EXT2 P1 2020 HSC 13c

i. Strategy 1

Using Pythagoras:

$$\begin{aligned} x &= \sqrt{(a+b)^2 - (a-b)^2} \\ &= \sqrt{4ab} \\ &= 2\sqrt{ab} \end{aligned}$$

$a + b$ is a hypotenuse

$$a + b \geq x$$

$$a + b \geq 2\sqrt{ab}$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Strategy 2

$$(\sqrt{a} - \sqrt{b})^2 \geq 0$$

$$a - 2\sqrt{a}\sqrt{b} + b \geq 0$$

$$a + b \geq 2\sqrt{ab}$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

ii. Let $a = p$, $b = 2q$

Using part i:

$$\frac{p+2q}{2} \geq \sqrt{2pq}$$

$$p+2q \geq 2\sqrt{2pq}$$

$$p^2 + 4pq + 4q^2 \geq 8pq$$

$$p^2 + 4q^2 \geq 4pq$$

7. Proof, EXT2 P1 2023 HSC 12b

$$\begin{aligned}(x-y)^2 &\geq 0 \\ x^2 - 2xy + y^2 &\geq 0 \\ x^2 + y^2 &\geq 2xy \\ \frac{2xy}{x^2 + y^2} &\leq 1 \dots (1)\end{aligned}$$

$$\begin{aligned}\frac{(x+y)^2}{x^2+y^2} &= \frac{x^2+2xy+y^2}{x^2+y^2} \\ &= \frac{x^2+y^2}{x^2+y^2} + \underbrace{\frac{2xy}{x^2+y^2}}_{\text{see (1) above}} \\ &\leq 1+1 \\ &\leq 2\end{aligned}$$

8. Proof, EXT2 P1 2006 HSC 8a

$$\begin{aligned}\text{i.} \quad 0 &\leq t \leq \frac{1}{\sqrt{2}} \\ 0 &\leq t^2 \leq \frac{1}{2} \\ 0 &\geq -t^2 \geq -\frac{1}{2} \\ 1 &\geq 1-t^2 \geq \frac{1}{2} \\ 1 &\leq \frac{1}{1-t^2} \leq 2 \\ 2t^2 &\leq \frac{2t^2}{1-t^2} \leq 4t^2, \quad (2t^2 > 0) \\ \therefore 0 &\leq \frac{2t^2}{1-t^2} \leq 4t^2\end{aligned}$$

$$\begin{aligned}\text{ii.} \quad \frac{1}{1+t} + \frac{1}{1-t} - 2 &= \frac{(1-t) + (1+t) - 2(1-t^2)}{1-t^2} \\ &= \frac{2t^2}{1-t^2}\end{aligned}$$

Substituting into part (i)

$$\therefore 0 \leq \frac{1}{1+t} + \frac{1}{1-t} - 2 \leq 4t^2$$

$$\begin{aligned}\text{iii.} \quad \int_0^x 0 \, dt &\leq \int_0^x \left(\frac{1}{1+t} + \frac{1}{1-t} - 2 \right) dt \leq \int_0^x 4t^2 \, dt \\ 0 &\leq [\log_e(1+t) - \log_e(1-t) - 2t]_0^x \leq \left[\frac{4t^3}{3} \right]_0^x \\ 0 &\leq [\log_e(1+x) - \log_e(1-x) - 2x] \leq \left[\frac{4x^3}{3} \right] \\ 0 &\leq \log_e \left(\frac{1+x}{1-x} \right) - 2x \leq \frac{4x^3}{3}\end{aligned}$$

iv. Since $e^a > e^b$ when $a > b$

$$\begin{aligned}e^0 &\leq e^{[\ln(\frac{1+x}{1-x}) - 2x]} \leq e^{\frac{4x^3}{3}} \\ 1 &\leq e^{\ln(\frac{1+x}{1-x})} \times e^{-2x} \leq e^{\frac{4x^3}{3}} \\ 1 &\leq \left(\frac{1+x}{1-x} \right) e^{-2x} \leq e^{\frac{4x^3}{3}}\end{aligned}$$

9. Proof, EXT2 P1 2015 HSC 15c

i. Since $(x-y)^2 = x^2 + y^2 - 2xy$

and $(x-y)^2 \geq 0$

$$0 \leq x^2 + y^2 - 2xy$$

$$2xy \leq x^2 + y^2$$

$$\therefore \sqrt{xy} \leq \sqrt{\frac{x^2 + y^2}{2}}$$

ii. $\sqrt{ab} \leq \sqrt{\frac{a^2 + b^2}{2}}$, (part (i))

$$\sqrt{cd} \leq \sqrt{\frac{c^2 + d^2}{2}}, \quad (\text{part (i)})$$

$$\begin{aligned} \sqrt{ab} \sqrt{cd} &\leq \sqrt{\frac{a^2 + b^2}{2}} \cdot \sqrt{\frac{c^2 + d^2}{2}} \\ &\leq \sqrt{\left(\frac{a^2 + b^2}{2}\right) \cdot \left(\frac{c^2 + d^2}{2}\right)} \end{aligned}$$

$$\sqrt{xy} \leq \frac{x+y}{2} \quad (\text{given}):$$

$$\sqrt{ab} \sqrt{cd} \leq \frac{1}{2} \left(\frac{a^2 + b^2 + c^2 + d^2}{2} \right)$$

$$\sqrt{abcd} \leq \frac{a^2 + b^2 + c^2 + d^2}{4}$$

$$\therefore \sqrt[4]{abcd} \leq \sqrt{\frac{a^2 + b^2 + c^2 + d^2}{4}}$$

♦♦ Mean mark 29%.

10. Proof, EXT2 P1 SM-Bank 6

i. $x^2 + y^2 + z^2 - yz - zx - xy > 0$

Multiply $\times 2$

$$2x^2 + 2y^2 + 2z^2 - 2yz - 2zx - 2xy > 0$$

$$(x^2 - 2xy + y^2) + (y^2 - 2yz + z^2) + (z^2 - 2zx + x^2) > 0$$

$$(x-y)^2 + (y-z)^2 + (z-x)^2 > 0$$

Square of any rational number > 0

$\therefore$ Statement is true.

ii. $(x+y+z)^2 = x^2 + xy + xz + yx + y^2 + yz + zx + zy + z^2$

$$1 = x^2 + y^2 + z^2 + 2xy + 2yz + 2xz$$

Consider statement in part (i):

$$x^2 + y^2 + z^2 - yz - zx - xy > 0$$

$$\Rightarrow (x+y+z)^2 - (x^2 + y^2 + z^2 - yz - zx - xy) < 1$$

$$3xy + 3yz + 3zx < 1$$

$$\therefore yz + zx + xy < \frac{1}{3}$$

11. Proof, EXT2 P1 2011 HSC 5b

If $p+q \geq r$, then $p+q-r \geq 0$

♦ Mean mark 41%.

$$\begin{aligned} \text{LHS} &= \frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \\ &= \frac{p(1+q)(1+r) + q(1+p)(1+r) - r(1+p)(1+q)}{(1+p)(1+q)(1+r)} \\ &= \frac{p(1+q+r+qr) + q(1+p+r+pr) - r(1+p+q+pq)}{(1+p)(1+q)(1+r)} \\ &= \frac{p+pq+pr+pqr+q+pq+qr+pqr-r-pr-qr-pqr}{(1+p)(1+q)(1+r)} \\ &= \frac{(p+q-r) + pq(2+r)}{(1+p)(1+q)(1+r)} \\ &\geq \frac{pq(2+r)}{(1+p)(1+q)(1+r)} \quad (\text{Since } p+q-r \geq 0) \\ &\geq 0 \quad (\text{Since } p > 0, q > 0, r > 0) \end{aligned}$$

12. Proof, EXT2 P1 2016 HSC 14c

Show $x\sqrt{x} + 1 \geq x + \sqrt{x}$, for $x \geq 0$

$$\begin{aligned} (\text{LHS})^2 &= x^3 + 2x\sqrt{x} + 1 \\ &= 2x\sqrt{x} + (x+1)(x^2-x+1) \end{aligned}$$

♦ Mean mark 39%.

$$(x-1)^2 \geq 0$$

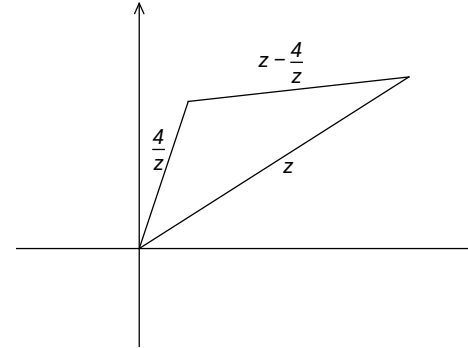
$$x^2 - 2x + 1 \geq 0$$

$$\therefore x^2 + 1 \geq 2x$$

$$\begin{aligned} \therefore (\text{LHS})^2 &\geq 2x\sqrt{x} + (x+1)(2x-x) \\ &\geq 2x\sqrt{x} + x(x+1) \\ &\geq x^2 + 2x\sqrt{x} + x \\ &\geq (x + \sqrt{x})^2 \\ &\geq (\text{RHS})^2 \end{aligned}$$

$$\therefore \text{LHS} \geq \text{RHS for } x \geq 0$$

13. Proof, EXT2 P1 2022 HSC 15d



♦♦♦ Mean mark 27%.

Triangle Inequality: $|x| + |y| \geq |x + y|$

$$|z| \leq \left| z - \frac{z}{4} \right| + \left| \frac{4}{z} \right|$$

$$|z| \leq 2 + \frac{4}{|z|} \quad \left(\text{using } \left| z - \frac{z}{4} \right| = 2 \right)$$

$$|z|^2 \leq 2|z| + 4$$

$$|z|^2 - 2|z| - 4 \leq 0$$

$$|z| \leq \frac{2 + \sqrt{2^2 + 4 \times 4}}{2} \quad (|z| \geq 0)$$

$$|z| \leq \frac{2 + \sqrt{20}}{2}$$

$$|z| \leq 1 + \sqrt{5} \quad \dots \text{ as required}$$

14. Proof, EXT2 P1 2023 HSC 16b

i. Prove $x > \ln x$ for $x > 0$:

$\Rightarrow$ Show $f(x) = x - \ln x > 0$

SP's occur when $f'(x) = 1 - \frac{1}{x} = 0$

SP at $(1, 1)$

$f'' = x^{-2} > 0, \forall x > 0$

SP at $(1, 1)$ is a global minimum for $x > 0$

$\Rightarrow f(x) \geq 1 > 0$

$\therefore x > \ln x$ for $x > 0$

ii. $x > \ln x$ for $x > 0 \Rightarrow e^x > x$ (by definition)

Choose any positive integer n :

$e^n > n$

$e^{n-1} > n - 1$

$\vdots$

$e^2 > 2$

$e^1 > 1$

Multiply each side of the equations above:

$e^n \times e^{n-1} \times \dots \times e^1 > n(n-1)(n-2)\dots(2)(1)$

$e^{n+(n-1)+(n-2)+\dots+2+1} > n!$

$e^{\frac{n(n+1)}{2}} > n!$ (using AP formula $S_n = \frac{n}{2}(a+l)$)

$e^{\frac{n(n+1)}{2} \times 2} > (n!)^2$

$e^{n^2+n} > (n!)^2 \dots$ as required

♦ Mean mark (i) 39%.

♦♦ Mean mark (ii) 28%.

15. Proof, EXT2 P1 2007 HSC 7a

i. Let $g(x) = \sin x - x$

$g'(x) = \cos x - 1 \leq 1$ for all $x > 0$

$\Rightarrow g(x)$ is a decreasing function

When $x = 0, g(0) = 0$

Considering $g(x)$ when $x > 0$,

$g(x) < 0$

$\sin x - x < 0$

$\sin x < x$ for all $x > 0$

ii. $f(x) = \sin x - x + \frac{x^3}{6}$

$f'(x) = \cos x - 1 + \frac{x^2}{2}$

$f''(x) = x - \sin x$

$\therefore f''(x) > 0$ (using part (i))

$\therefore f(x)$ is concave up for $x > 0$.

iii. $f''(x) > 0$ (part (ii))

$\Rightarrow f'(x)$ is an increasing function

When $x = 0, f'(0) = 0$

$\Rightarrow f'(x) > 0$ for $x > 0$

$\therefore f(x)$ has a positive gradient that steepens

for $x > 0$, and $f(0) = 0$

$f(x) > 0$

$\sin x - x + \frac{x^3}{6} > 0$

$\therefore \sin x > x - \frac{x^3}{6}$ for $x > 0$

MARKER'S COMMENT: A geometric proof using arc length and a right angled triangle caused problems as few students could deal with the case $x > \pi$.

16. Proof, EXT2 P1 2012 HSC 15a

i. $(\sqrt{a} - \sqrt{b})^2 \geq 0$

$$a - 2\sqrt{ab} + b \geq 0$$

$$a + b \geq 2\sqrt{ab}$$

$$\sqrt{ab} \leq \frac{a+b}{2}$$

ii. Solution 1

Since $1 \leq x \leq y$

$$y - x \geq 0$$

$$y(x-1) - x(x-1) \geq 0, \quad (x-1 \geq 0)$$

$$xy - x^2 + x - y \geq 0$$

$$\therefore xy - x^2 + x \geq y$$

Solution 2

$$x(y-x+1) = xy - x^2 + x$$

$$= -y + xy - x^2 + x + y$$

$$= y(x-1) - x(x-1) + y$$

$$= (x-1)(y-x) + y$$

Since $x-1 \geq 0$ and $y-x \geq 0$

$$\Rightarrow (x-1)(y-x) + y \geq y$$

$$\therefore x(y-x+1) \geq y$$

iii. Let $c = n - j + 1$, $c > 0$ as $n \geq j$.

$$\Rightarrow \sqrt{j(n-j+1)} \text{ can be written as } \sqrt{jc}.$$

$$\sqrt{jc} \leq \frac{j+c}{2} \quad (\text{part (i)})$$

$$\sqrt{j(n-j+1)} \leq \frac{j+n-j+1}{2}$$

$$\Rightarrow \sqrt{j(n-j+1)} \leq \frac{n+1}{2}$$

$$j(n-j+1) \geq n \quad (\text{part (ii)})$$

$$\Rightarrow \sqrt{j(n-j+1)} \geq \sqrt{n}$$

$$\therefore \sqrt{n} \leq \sqrt{j(n-j+1)} \leq \frac{n+1}{2}$$

iv. From (iii) $n \leq j(n-j+1) \leq \frac{n+1}{2}$

Let j take on the values from 1 to n .

$$j=1: \sqrt{n} \leq \sqrt{1(n)} \leq \frac{n+1}{2}$$

$$j=2: \sqrt{n} \leq \sqrt{2(n-1)} \leq \frac{n+1}{2}$$

$\vdots$

$$j=n: \sqrt{n} \leq \sqrt{n(1)} \leq \frac{n+1}{2}$$

Multiply the corresponding parts of each line

$$(\sqrt{n})^n \leq \sqrt{n! \times n!} \leq \left(\frac{n+1}{2}\right)^n$$

$$(\sqrt{n})^n \leq n! \leq \left(\frac{n+1}{2}\right)^n$$

♦♦ Mean mark part (ii) 28%.

♦ Mean mark part (iii) 39%.

♦♦♦ Mean mark part (iv) just 2%!

17. Proof, EXT2 P1 2015 HSC 15b

i. $1 - x^2 \leq 1 \quad \text{for } x \geq 0$

$$(1 - x)(1 + x) \leq 1$$
$$(1 - x) \leq \frac{1}{1 + x}$$

Since $1 + x \geq 1$ when $x \geq 0$

$$\Rightarrow \frac{1}{1 + x} \leq 1$$
$$\therefore 1 - x \leq \frac{1}{1 + x} \leq 1.$$

ii. $\int_0^{\frac{1}{n}} (1 - x) \, dx \leq \int_0^{\frac{1}{n}} \frac{dx}{1 + x} \leq \int_0^{\frac{1}{n}} 1 \, dx$

$$\left[x - \frac{x^2}{2} \right]_0^{\frac{1}{n}} \leq [\ln(1 + x)]_0^{\frac{1}{n}} \leq [x]_0^{\frac{1}{n}}$$
$$\frac{1}{n} - \frac{1}{2n^2} \leq \ln\left(1 + \frac{1}{n}\right) \leq \frac{1}{n}$$
$$1 - \frac{1}{2n} \leq n \ln\left(1 + \frac{1}{n}\right) \leq 1, \quad (n \geq 1)$$

iii. $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2n}\right) \leq \lim_{n \rightarrow \infty} \left\{ n \ln\left(1 + \frac{1}{n}\right) \right\} \leq \lim_{n \rightarrow \infty} (1)$

$$1 \leq \lim_{n \rightarrow \infty} \left\{ n \ln\left(1 + \frac{1}{n}\right) \right\} \leq 1$$
$$\therefore \lim_{n \rightarrow \infty} \left(n \ln\left(1 + \frac{1}{n}\right) \right) = 1$$
$$\lim_{n \rightarrow \infty} \ln\left(1 + \frac{1}{n}\right)^n = 1$$
$$\therefore \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

♦♦ Mean mark 21%.
STRATEGY: The conversion of the middle term from a fraction into a logarithm should flag the need for integration of each term.

♦♦ Mean mark 29%.

18. Vectors, EXT2 V1 2021 HSC 16a

i. **Triangle inequality:** $|x| + |y| \geq |x + y|$

$$|x| + |y| + |z| = \left| \underset{\sim}{x}_i \right| + \left| \underset{\sim}{y}_j \right| + \left| \underset{\sim}{z}_k \right|$$
$$\geq \left| \underset{\sim}{x}_i + \underset{\sim}{y}_j \right| + \left| \underset{\sim}{z}_k \right|$$
$$\geq \left| \underset{\sim}{x}_i + \underset{\sim}{y}_j + \underset{\sim}{z}_k \right|$$
$$\geq 1 \quad \left(\left| \overrightarrow{OP} \right| = \left| \underset{\sim}{x}_i + \underset{\sim}{y}_j + \underset{\sim}{z}_k \right| = 1 \right)$$

♦ Mean mark (i) 50%.

ii. Using the dot product:

$$\underset{\sim}{a} \cdot \underset{\sim}{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$
$$\underset{\sim}{a} \cdot \underset{\sim}{b} = \left| \underset{\sim}{a} \right| \left| \underset{\sim}{b} \right| \cos \theta$$

♦ Mean mark (ii) 42%.

$$a_1 b_1 + a_2 b_2 + a_3 b_3 = \sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2} \cdot \cos \theta$$
$$|a_1 b_1 + a_2 b_2 + a_3 b_3| = \sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2} \cdot |\cos \theta|$$

Since $-1 \leq \cos \theta \leq 1 \Rightarrow |\cos \theta| \leq 1$

$$\therefore |a_1 b_1 + a_2 b_2 + a_3 b_3| \leq \sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2}$$

iii. Using part (ii) with vectors:

$$\underset{\sim}{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \underset{\sim}{b} = \begin{pmatrix} |x| \\ |y| \\ |z| \end{pmatrix}$$

$$| |x| + |y| + |z| | \leq \sqrt{1^2 + 1^2 + 1^2} \cdot \sqrt{x^2 + y^2 + z^2}$$
$$|x| + |y| + |z| \leq \sqrt{3}$$

♦♦♦ Mean mark (iii) 14%.

19. Proof, EXT2 P1 2012 HSC 16c

i. Let $P(1)$ = (After 1st number chosen, the second draw matches)

$$P(1) = \frac{n}{n} \times \frac{1}{n} = \frac{1}{n}$$

Let $P(2)$ = (After 1st two numbers chosen, the third draw matches)

$$P(2) = \frac{n}{n} \times \frac{n-1}{n} \times \frac{2}{n}$$

$$P(3) = \frac{n}{n} \times \frac{n-1}{n} \times \frac{n-2}{n} \times \frac{3}{n}$$

$\vdots$

$\Rightarrow$ On the $(k+1)$ th draw

$$\begin{aligned} P(k) &= \frac{n}{n} \times \frac{n-1}{n} \times \frac{n-2}{n} \times \dots \times \frac{n-k+1}{n} \times \frac{k}{n} \\ &= \frac{(n-1)!}{n^k} \times \frac{k}{1 \times 2 \times \dots \times (n-k)} \\ &= \frac{(n-1)! k}{n^k (n-k)!} \end{aligned}$$

◆◆◆ Mean mark part (i) 7%.

ii.

$$P(k) \geq P(k-1)$$

$$\frac{(n-1)! k}{n^k (n-k)!} \geq \frac{(n-1)! (k-1)}{n^{k-1} (n-k+1)!}$$

$$k(n-k+1) \geq n(k-1)$$

$$kn - k^2 + k \geq nk - n$$

$$\therefore k^2 - k - n \leq 0$$

◆ Mean mark part (ii) 38%.

iii. $\sqrt{n + \frac{1}{4}} > k - \frac{1}{2}$

$$n + \frac{1}{4} > \left(k - \frac{1}{2}\right)^2$$

$$n + \frac{1}{4} > k^2 - k + \frac{1}{4}$$

$$n > k^2 - k$$

$$n > k(k-1)$$

Since n and k are integers such that $1 \leq k \leq n$.

$\Rightarrow n$ is at least the next integer after k .

$\Rightarrow (k-1)$ is the integer before k .

$$\therefore n > k^2 - k + \frac{1}{4}$$

n

$$> \left(k - \frac{1}{2}\right)^2$$

$$\therefore \sqrt{n} > k - \frac{1}{2}$$

iv. Find the largest integer k for which $P(k) \geq P(k-1)$

$$P(k) \geq P(k-1) \text{ when } k^2 - k - n \leq 0 \text{ (part (ii))}$$

Solving $k^2 - k - n \leq 0$

$$\frac{1 - \sqrt{4n+1}}{2} \leq k \leq \frac{1 + \sqrt{4n+1}}{2}$$

Given $n > 0$, $\frac{1 - \sqrt{4n+1}}{2} < 0$

$$\therefore 1 \leq k \leq \frac{1 + \sqrt{4n+1}}{2}$$

Since $4n+1$ is not a perfect square and k is an integer

$$k < \frac{1 + \sqrt{1+4n}}{2}$$

$$k < \frac{1}{2} + \sqrt{n + \frac{1}{4}}$$

$$k - \frac{1}{2} < \sqrt{n + \frac{1}{4}}$$

$$k - \frac{1}{2} < \sqrt{n} \quad (\text{from part (iii)})$$

$$k < \frac{1}{2} + \sqrt{n}$$

$\therefore P(k)$ is greatest when k is the closest integer to n .

◆◆◆ Mean mark part (iii) 1%.

20. Proof, EXT2 P1 2013 HSC 16a

i. $P(x) = 2x^3 - 15x^2 + 24x + 16, \quad x \geq 0$

$$P'(x) = 6x^2 - 30x + 24$$
$$= 6(x^2 - 5x + 4)$$
$$= 6(x-1)(x-4)$$
$$P''(x) = 12x - 30$$

MAX or MIN when $P'(x) = 0$
i.e. when $x = 1$ or 4
 $P''(1) = -6 < 0$ $P''(4) = 18 > 0$
 $\therefore$ Minimum turning point of $P(x)$ at $x = 4$,
 $P(4) = 128 - 240 + 96 + 16 = 0$

Checking limits:
 $P(0) = 16$
As $x \rightarrow \infty, y \rightarrow \infty$
 $\therefore$ Minimum value of $P(x)$ is 0 when $x = 4$.

ii. LHS $= (x+1)(x^2 + (x+4)^2)$

$$= (x+1)(2x^2 + 8x + 16)$$
$$= 2x^3 + 8x^2 + 16x + 2x^2 + 8x + 16$$
$$= 2x^3 + 10x^2 + 24x + 16$$
$$= (2x^3 - 15x^2 + 24x + 16) + 25x^2$$
$$= P(x) + 25x^2$$

The minimum value of $P(x)$ for $x \geq 0$ is 0 ,

$$\Rightarrow P(x) + 25x^2 \geq 25x^2$$

$\therefore (x+1)(x^2 + (x+4)^2) \geq 25x^2, \quad \text{for } x \geq 0$

iii. $(m+n)^2 + (m+n+4)^2$

$$= (m+n)^2 + (m+n)^2 + 8(m+n) + 16$$
$$= 2(m+n)^2 + 8(m+n) + 16$$

Let $x = m+n$
 $(m+n)^2 + (m+n+4)^2 = 2x^2 + 8x + 16$
Using part (ii)
 $(x+1)(2x^2 + 8x + 16) \geq 25x^2$

♦ Mean mark 43%.
STRATEGY: The best responses carefully considered the domain restrictions, notably $P(0)$.

♦ Mean mark 47%.

♦♦ Mean mark 19%.
STRATEGY: The best responses carefully considered the domain restrictions, notably $P(0)$.

$$2x^2 + 8x + 16 \geq \frac{25x^2}{x+1}, \quad x \geq 0$$

$$(m+n)^2 + (m+n+4)^2 \geq \frac{25(m+n)^2}{m+n+1}$$

Consider $(m-n)^2$

$$\text{Since } (m-n)^2 \geq 0$$

$$m^2 + n^2 \geq 2mn$$

$$(m+n)^2 - 2mn \geq 2mn$$

$$(m+n)^2 \geq 4mn$$

$$\therefore (m+n)^2 + (m+n+4)^2 \geq \frac{100mn}{m+n+1}$$

21. Proof, EXT2 P1 2021 HSC 15a

i. Show $\sqrt{abc} \leq \frac{a^2 + b^2 + 2c}{4}$

$$\sqrt{abc} = \sqrt{(ab)c} \leq \frac{ab + c}{2} \dots (1)$$

Let $x = a^2$ and $y = b^2$:

$$\sqrt{xy} = \sqrt{a^2b^2} \leq \frac{a^2 + b^2}{2}$$
$$ab \leq \frac{a^2 + b^2}{2} \dots (2)$$

Substitute (2) into (1):

$$\sqrt{abc} \leq \frac{\frac{a^2 + b^2}{2} + c}{2}$$
$$\sqrt{abc} \leq \frac{a^2 + b^2 + 2c}{4}$$

ii. Show $\sqrt{abc} \leq \frac{a^2 + b^2 + c^2 + a + b + c}{6}$

Similarly (to part a):

If $x = b^2$ and $y = c^2$

$$\sqrt{abc} \leq \frac{b^2 + c^2 + 2a}{4}$$

If $x = c^2$ and $y = a^2$

$$\sqrt{abc} \leq \frac{c^2 + a^2 + 2b}{4}$$

$$\therefore 3\sqrt{abc} \leq \frac{a^2 + b^2 + 2c + b^2 + c^2 + 2a + c^2 + a^2 + 2b}{4}$$

$$3\sqrt{abc} \leq \frac{2(a^2 + b^2 + c^2 + a + b + c)}{4}$$

$$\sqrt{abc} \leq \frac{a^2 + b^2 + c^2 + a + b + c}{6}$$

♦ Mean mark part (i) 50%.

♦♦♦ Mean mark part (ii) 26%.

22. Proof, EXT2 P1 2021 HSC 15b

i. $t_n = \frac{n(n+1)}{2}$

Odd triangular numbers:

→ Let $n = 2k - 1$ for integers $k \geq 1$

$$t_{2k-1} = \frac{(2k-1)(2k-1+1)}{2}$$
$$= \frac{2k(2k-1)}{2}$$
$$= 2k^2 - k \text{ (definition of hexagonal numbers)}$$

∴ $t_1, t_3 \dots$ are also hexagonal numbers.

ii. $t_2, t_4, t_6, \dots$

→ Let $n = 2k$ for integers $k \geq 1$

$$t_{2k} = \frac{2k(2k+1)}{2}$$
$$= 2k^2 + k$$

Find values for k, n that satisfy:

$$2k^2 + k = 2n^2 - n$$

$$2k^2 - 2n^2 + k + n = 0$$

$$2(k-n)(k+n) + (k+n) = 0$$

$$(k+n)(2k-2n+1) = 0$$

$k = -n \Rightarrow$ no solution ($k, n \geq 1$)

$2k = 2n - 1 \Rightarrow$ no solution ($2k$ is even, $2n - 1$ is odd)

∴ $t_2, t_4, t_6, \dots$ are not hexagonal numbers.

♦ Mean mark part (i) 50%.

♦ Mean mark part (ii) 1%!

23. Proof, EXT2 P1 2022 HSC 16c

i. Show $abc \leq \left(\frac{S}{6}\right)^{\frac{3}{2}}$

$$S = 2(ab + bc + ac)$$

$$A = \frac{ab + bc + ac}{3}$$

Using given relationship, where $x_1 = ab, x_2 = bc, \dots$

$$\frac{ab \times bc \times ca}{\left(\frac{ab+bc+ac}{3}\right)^3} \leq 1$$

$$(ab \times bc \times ca) \leq \left(\frac{ab + bc + ac}{3}\right)^3$$

$$(abc)^2 \leq \left(\frac{2(ab + bc + ac)}{6}\right)^3$$

$$abc \leq \left(\frac{S}{6}\right)^{\frac{3}{2}}$$

ii. If prism is a cube $a = b = c$

$$\Rightarrow V = a^3, \quad S = 6a^2$$

$$\left(\frac{S}{6}\right)^{\frac{3}{2}} = \left(\frac{6a^2}{6}\right)^{\frac{3}{2}} = a^3$$

In the case of a cube:

$$V = \left(\frac{S}{6}\right)^{\frac{3}{2}}$$

Also, $V \leq \left(\frac{S}{6}\right)^{\frac{3}{2}}$ (using part (i))

$\therefore$ For rectangular prisms with a given surface area, a cube has the maximum volume.

◆◆◆ Mean mark (i) 26%.

◆◆◆ Mean mark (ii) 26%.

24. Proof, EXT2 P1 2014 HSC 15a

$$a + b + c = 1, \quad a \leq b \leq c$$

$$(a + b + c)^2 = 1$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ac = 1$$

$$\text{Since } a \leq b \Rightarrow a^2 \leq ab$$

$$\text{Since } b \leq c \Rightarrow b^2 \leq bc$$

$$\text{Since } a \leq c \Rightarrow a^2 \leq ac$$

$$\Rightarrow a^2 + b^2 + c^2 + 2a^2 + 2b^2 + 2a^2 \leq 1$$

$$\therefore 5a^2 + 3b^2 + c^2 \leq 1 \quad \dots \text{ as required}$$

◆◆ Mean mark 13%.